PREDICTING EARTHFLOW MOVEMENT BASED ON ML TECHNIQUES

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Guide

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Objectives

01

Understanding parameters involved with earthflow movements

02

Learning Data Structures algorithms

03

Developing ML algorithms for predicting soil movements based on data

04

Development of ensemble techniques for betterment of prediction

Progress Made(In last semester)

LEARNING PYTHON AND SPECIFIC LIBRARY(NUMPY,PANDAS)

DISCUSSED METHODOLOGY TO BE USED

DATA PRE-PROCESSING

RESEARCH ON PROJECT

Progress Made

PREPROCESSING AND ANALYZING THE DATA

LEARNED DATA STRUCTURES AND SPECIFIC LIBRARIES

APPLYING MACHINE LEARNING MODELS ON DATA

TRAINING AND TESTING OF THE MODEL

RESULT FINDING

Parameters involved in soil movements

Intrinsic feature of soil

- Type of soil
- Shape and Slope of soil layer
- Friction angle
- Degree of saturation
- Undrained strength
- Types of vegetation and root type
- Covered area
- Porosity
- Plasticity Index

Trigger Factors

- Precipitation
- Earthquakes
- Volcanic activity
- Cuts and excavation
- Deforestation
- Mechanical stresses

Progress Made: Data Preprocessing

- Data is obtained from the landslide sensor placed in Gharpa Mandi district of Himachal Pradesh, India.
- This landslide is located on the northern Himalayan region at latitudes 31° 46' 34.28" N and longitudes 76° 59' 2.05" E, at an altitude of 1044 meter.
- We got data of aggregated acceleration of soil collected using MPU6050 3 axis accelerometer stored at a duration of 10 minutes.
- Converted these acceleration into respective displacements using the following equation here “time (t)” is 1 millisecond.

$$S = ut + \frac{1}{2}at^2$$

- From displacement components we calculated resultant displacement, then cleaned data and used cumulative resultant displacement for model training and testing.

DateTime	Resultant
2019-07-26 12:45:26	0.000000
2019-07-26 12:55:52	0.000000
2019-07-26 13:06:17	0.000000
2019-07-26 13:16:43	0.000000
2019-07-26 13:27:09	0.000000
...	...
2019-10-06 00:47:24	1934.279787
2019-10-06 00:57:49	1934.279787
2019-10-06 01:08:15	1934.279787
2019-10-06 01:18:41	1934.279787
2019-10-06 01:29:07	1934.279787

2348 rows × 1 columns

Progress Made: Machine Learning Models

- **Persistence Model:**

  Persistence model uses last data; It does not take any trend or Previous history into account.

  The persistence algorithm uses the value at the previous time step (t-1) to predict the expected outcome at the next time step (t+1).Key Parameters: **RMSE** Root mean Square Error

- **Auto Regression Model(AR) :**

  Autoregression is a time series model that uses observations from previous time

  steps as input to a regression equation to predict the value at the next time step.

$$\boldsymbol{y = b_0 + \Sigma bi * Xi}$$

  Here i vary from 0 to L (Lags) and b_i is linear coefficient, X_i is I^{th} previous value

- **ARIMA: AutoRegressive Integrated Moving Average**

  ARIMA model can be understand by outlining each of its components

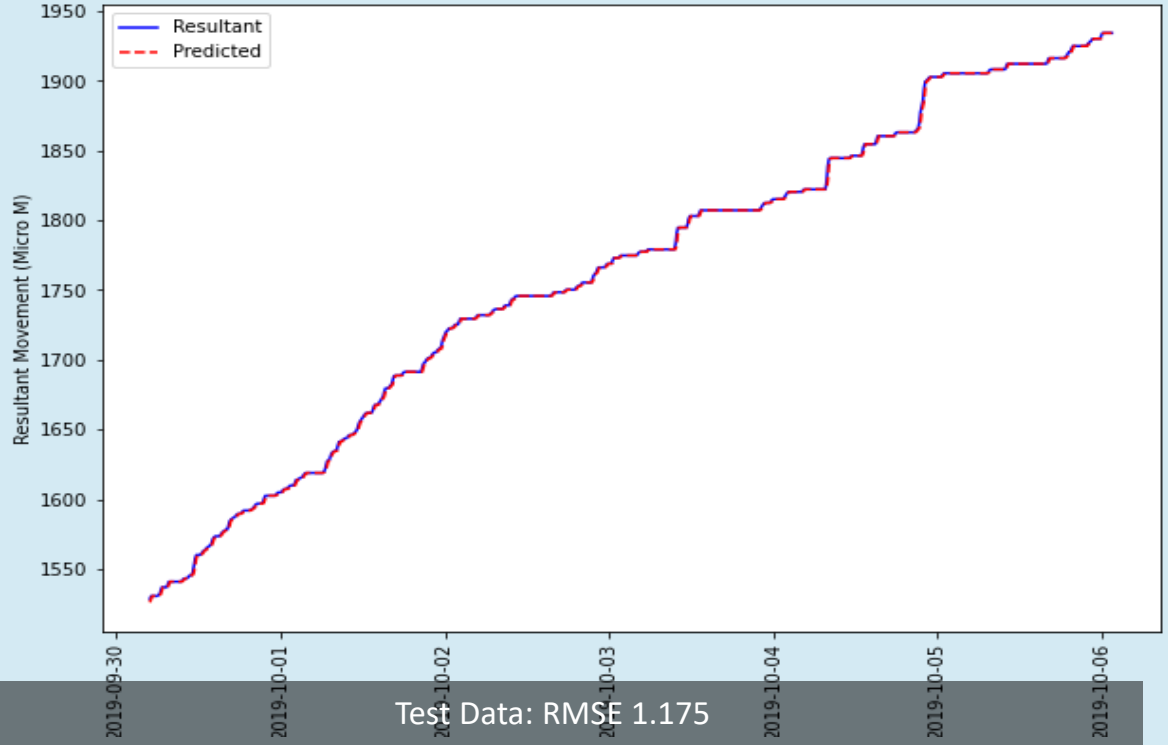
  Autoregression (AR) refers to a model that shows a changing variable that regresses on its own lagged, or prior, values.

  Integrated (I) represents the differencing of raw observations to allow for the time series to become stationary, i.e., data values are replaced by the difference between the data values and the previous values.

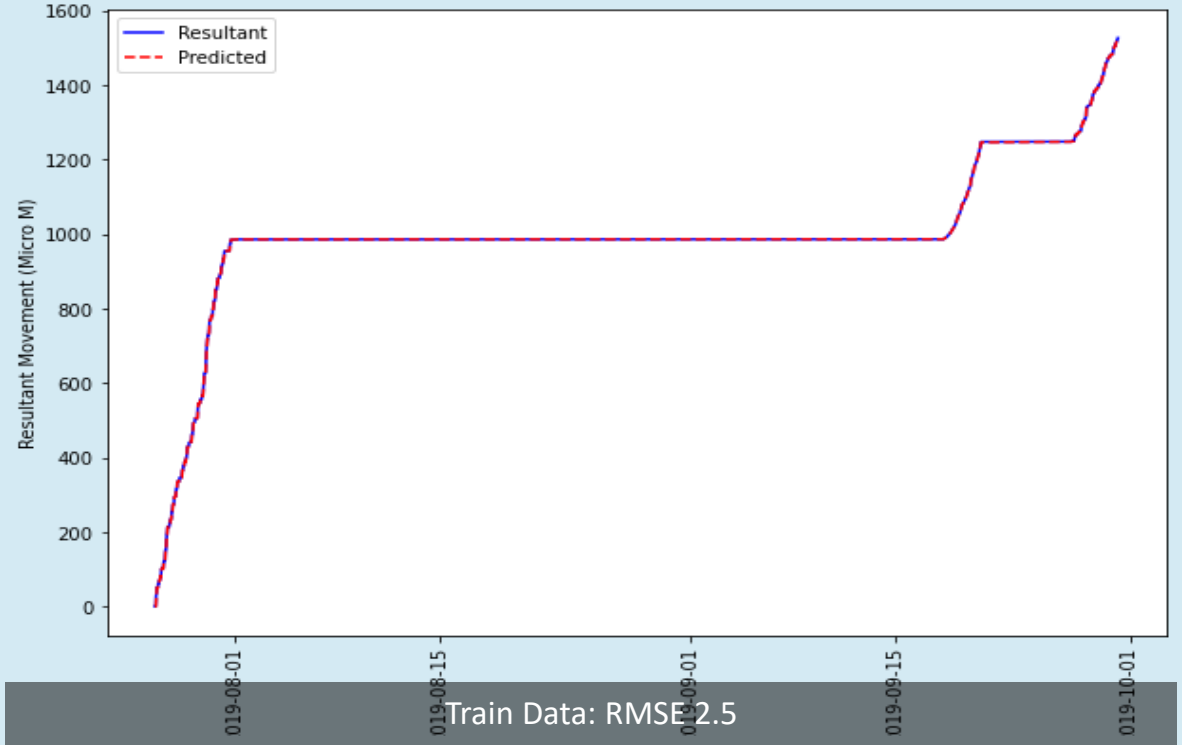
  Moving average (MA) incorporates the dependency between an observation and a residual error from a moving average model applied to lagged observations.

Results: Persistence Model

Persistence model uses last data, It does not take any trend or Previous history into account.



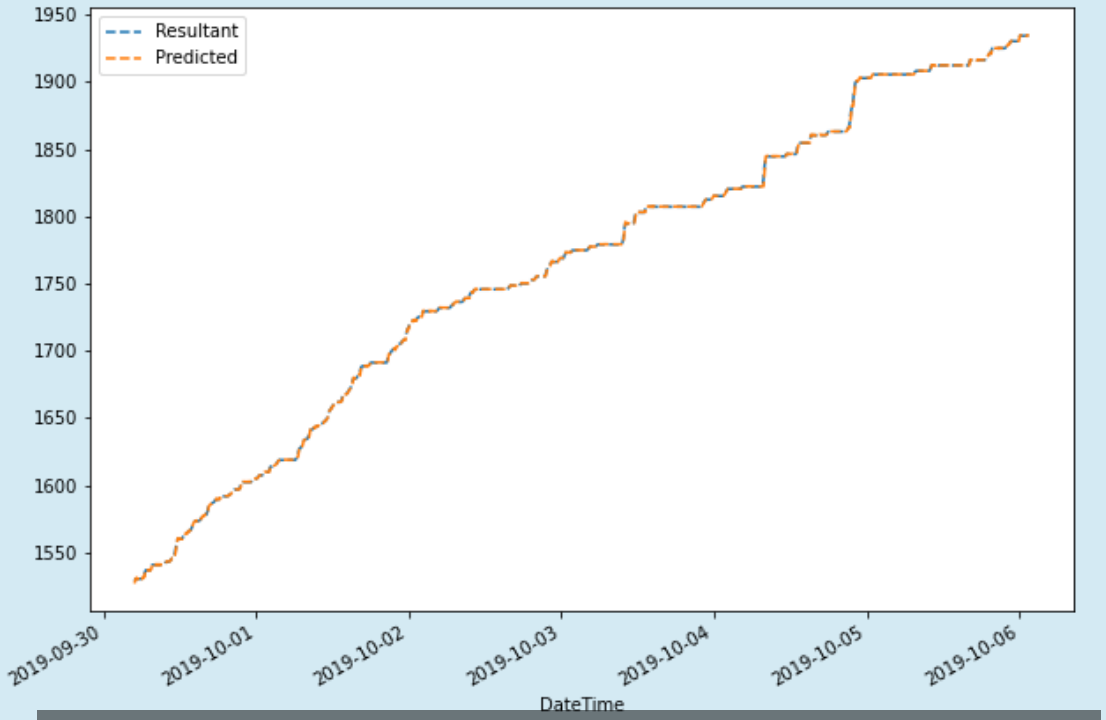
Test Data: RMSE 1.175



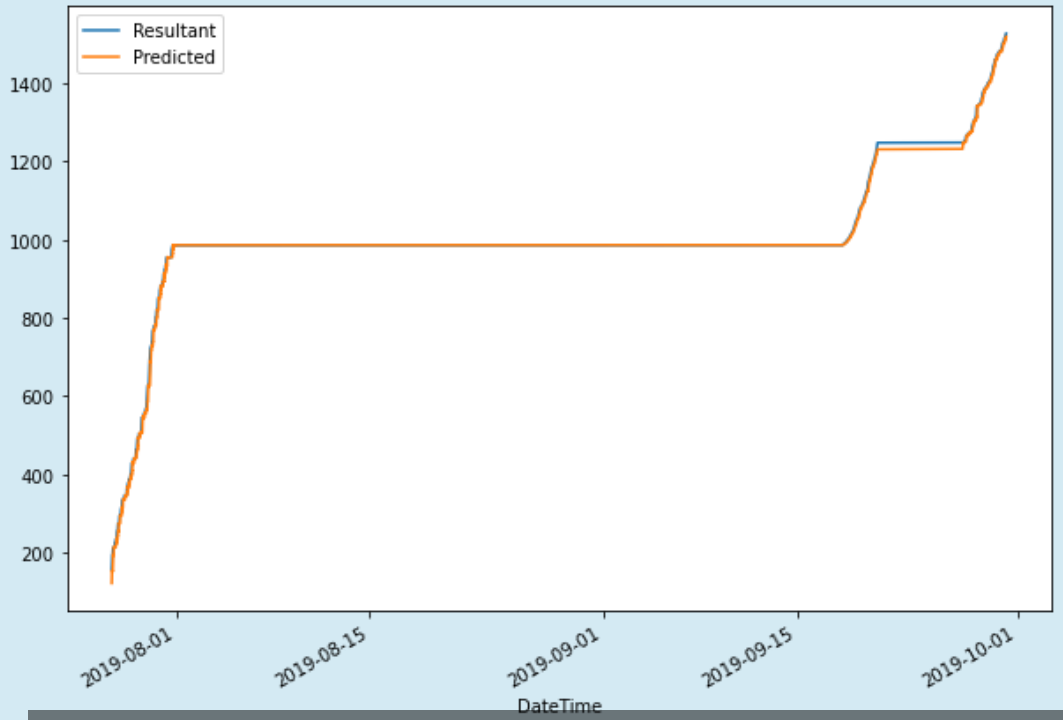
Train Data: RMSE 2.5

Results: Auto Regression

Autoregressive model predicts future behavior based on past behavior, the process is linear regression of the data in the current series against one or more past values in the same series.



Test Data: RMSE 0.44

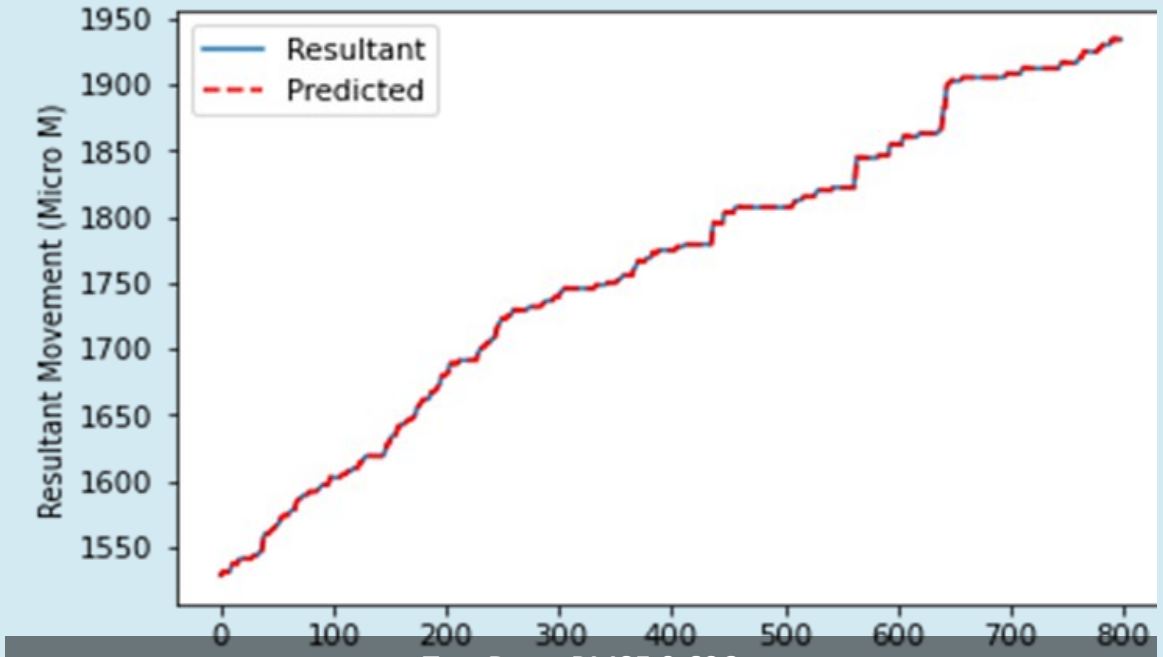


Train Data: RMSE 89

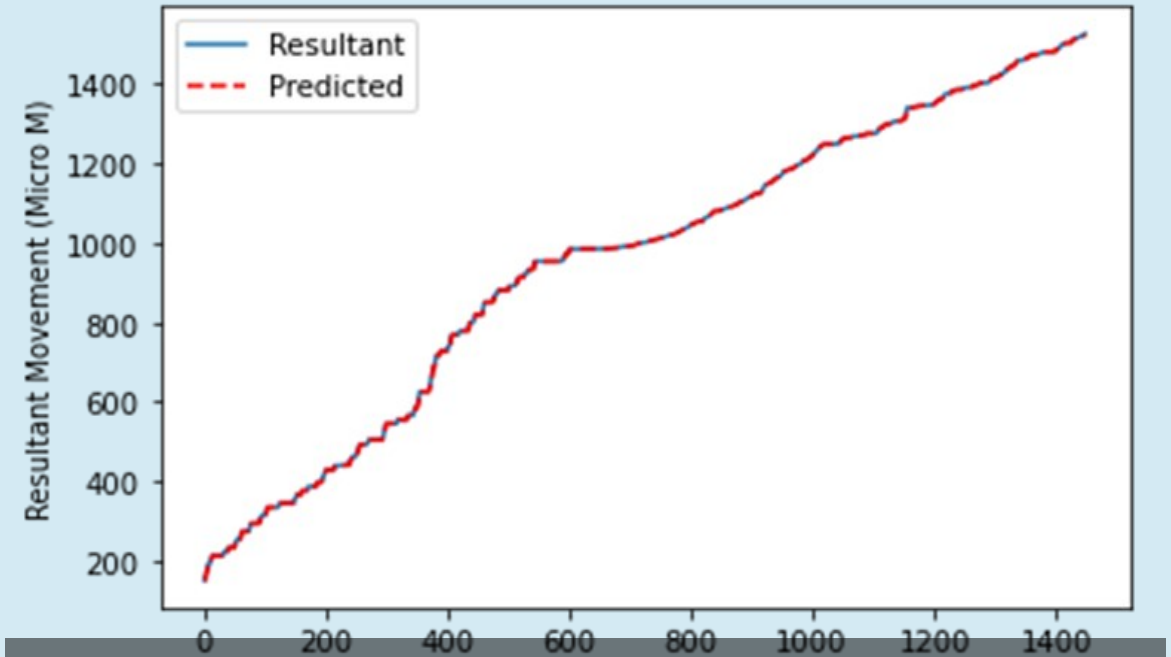
Results: ARIMA

AutoRegressive Integrated Moving Average, ARIMA(p,d,q) here p is order of Autoregressive part, d is degree of differencing involved, q order of moving average part.

ARIMA(5,1,0)



Test Data: RMSE 0.696



Train Data: RMSE 1.610

Results

Comparison

DateTime	Resultant	Persistence	Auto Regression	ARIMA
2019-09-30 22:16:20	1602.465411	1602.465411	1602.708140	1602.761120
2019-09-30 22:26:46	1602.465411	1602.465411	1602.459788	1602.289348
2019-09-30 22:37:11	1602.465411	1602.465411	1602.812568	1603.001157
2019-09-30 22:47:40	1602.465411	1602.465411	1602.414683	1602.465411
2019-09-30 22:58:03	1602.465411	1602.465411	1602.903549	1602.465411
...	...	...	...	...
2019-10-06 00:47:24	1934.279787	1934.279787	1934.053902	1933.875945
2019-10-06 00:57:49	1934.279787	1934.279787	1934.673952	1934.802006
2019-10-06 01:08:15	1934.279787	1934.279787	1933.998421	1934.279787
2019-10-06 01:18:41	1934.279787	1934.279787	1934.699521	1934.279787
2019-10-06 01:29:07	1934.279787	1934.279787	1934.160983	1934.279787

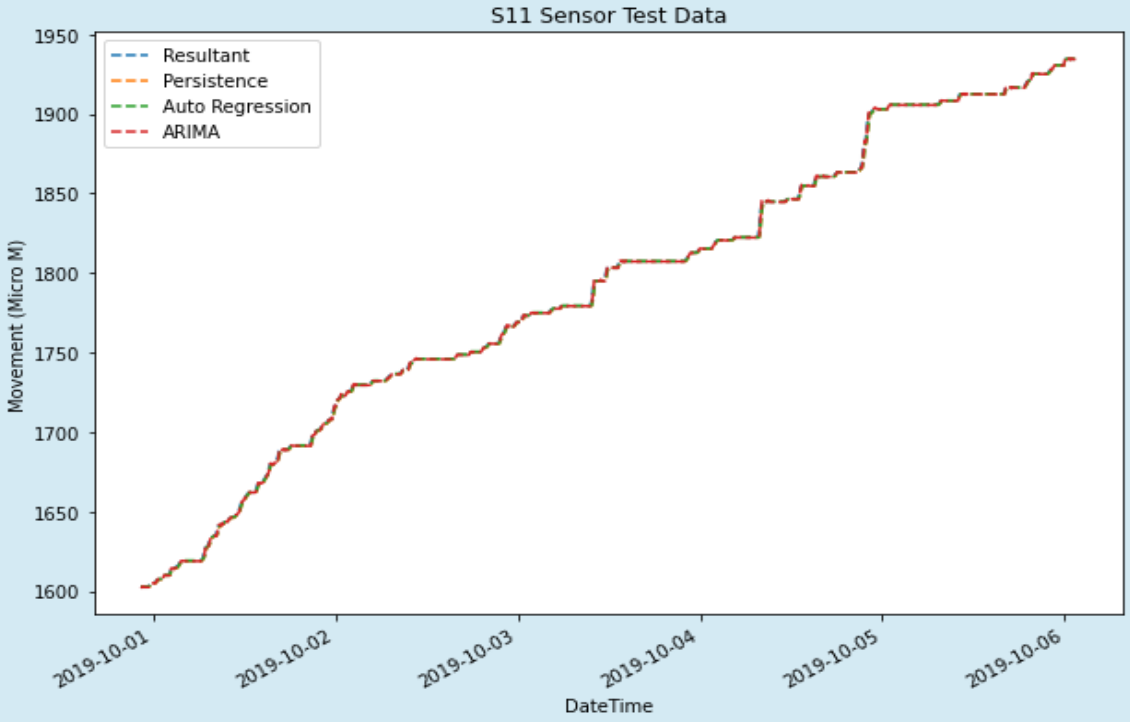
699 rows × 4 columns

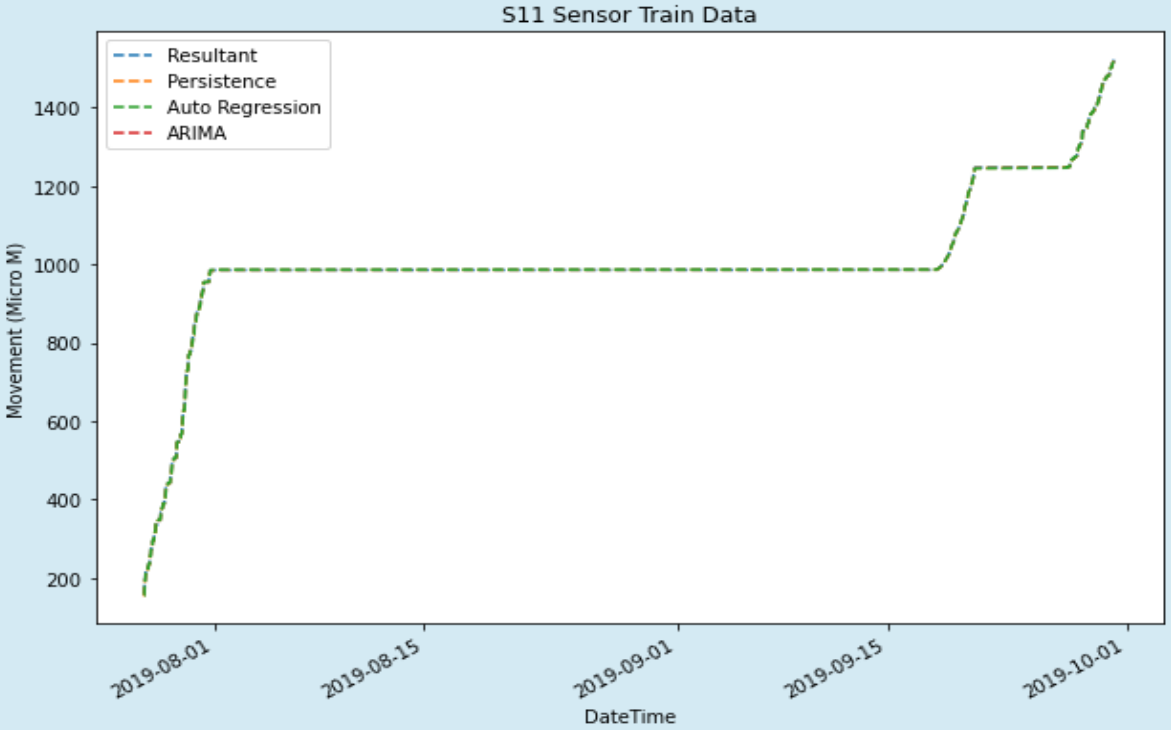
DateTime	Resultant	Persistence	Auto Regression	ARIMA
2019-07-27 07:00:21	157.376925	152.300250	122.448880	152.300250
2019-07-27 07:10:47	167.530275	157.376925	133.141504	157.386303
2019-07-27 07:21:12	172.606950	167.530275	143.946161	167.553888
2019-07-27 07:31:38	177.708150	172.606950	153.747307	172.624275
2019-07-27 07:42:04	187.886025	177.708150	153.665663	177.716801
...	...	...	...	...
2019-09-30 04:01:29	1520.193790	1520.193790	1514.983780	1520.194869
2019-09-30 04:11:55	1522.188146	1520.193790	1515.658855	1520.195334
2019-09-30 04:22:21	1524.187738	1522.188146	1516.125297	1522.196368
2019-09-30 04:32:46	1524.767768	1524.187738	1516.814995	1524.198886
2019-09-30 04:43:12	1525.930006	1524.767768	1518.261554	1524.774424

1448 rows × 4 columns

Results

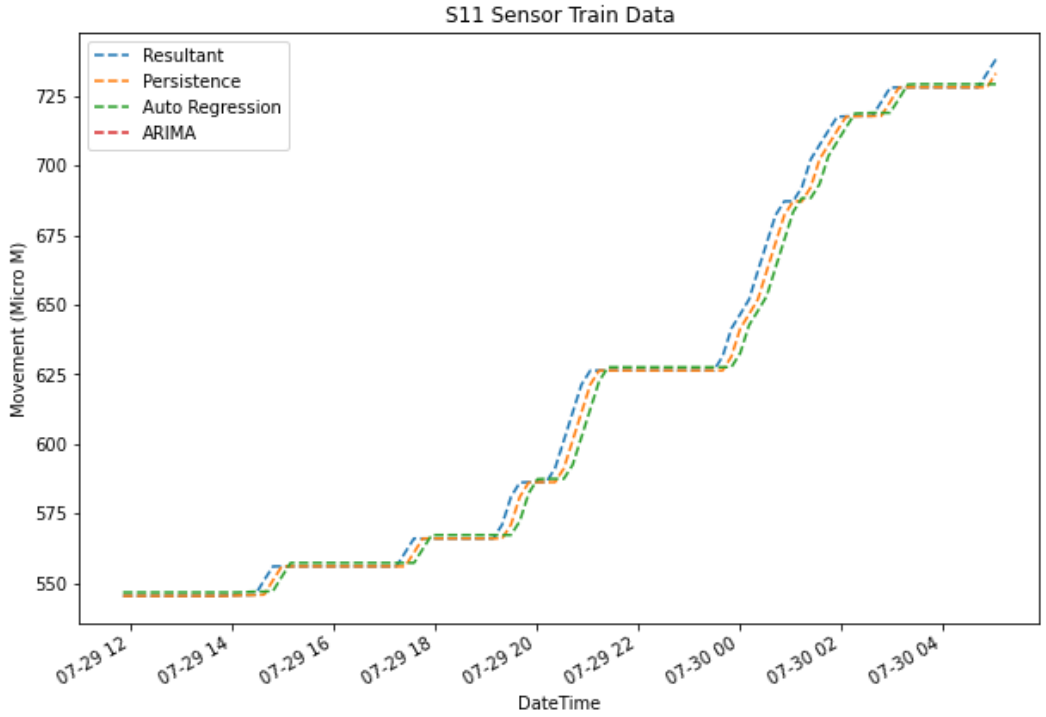
Comparison

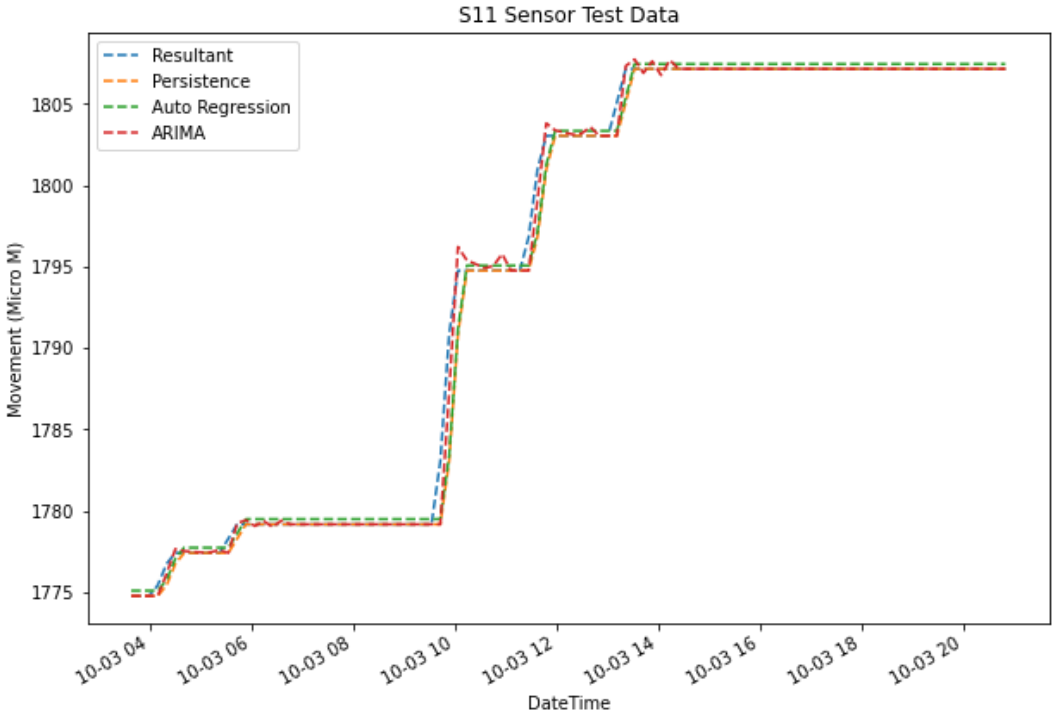




Results

Comparision





Proposed Time Plan

Week	Work
N/A	Collecting the data
1-10	Analyzing the data
2-5	Applying ML models on data
5-8	Testing and training of the model
5-8	Result finding
6-8	Comparing the results of each model
8-10	Explore other datasets
10-13	Develop novel ensembles for various models
14-15	Submit manuscripts for publication

Thank you

References :

Jason Brownlee, J. B. (2018). *Deep Learning for Time Series Forecasting*

https://machinelearningmastery.com/deep-learning-for-time-series-forecasting/

Ankush, Praveen, Priyanka, Akash (2020). Predictions of soil movements using Persistence, Auto- regression, and Neural network models: A case-study in Mandi, India